

Hydrodomus

Decentralizing hydrogen production.

Hydrogen made where it is used, from water and electricity

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Company pitch
Munich • Vienna Founders
Residency

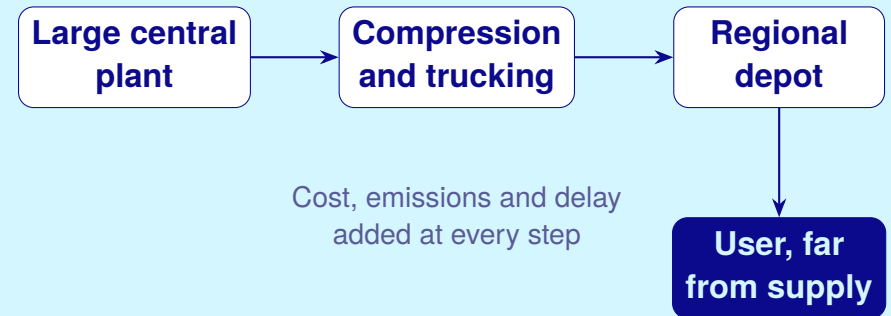
Hydrogen is made far away, then trucked in.

Most hydrogen today is made at large central plants, mostly from natural gas, then compressed and moved by truck or pipeline.

Every step adds cost, emissions and delay:

- **Expensive:** retail hydrogen runs roughly 15 to 30 USD/kg at the pump (industry estimates).
- **Not clean:** the centralized model inherits grey production.
- **Out of reach:** users far from a supplier are shut out.

It is the fossil fuel model again: centralized production, long logistics, dependent users.



Our answer: produce it at the point of use.



Illustration: centralized grey hydrogen versus clean, local production.

Today: centralized

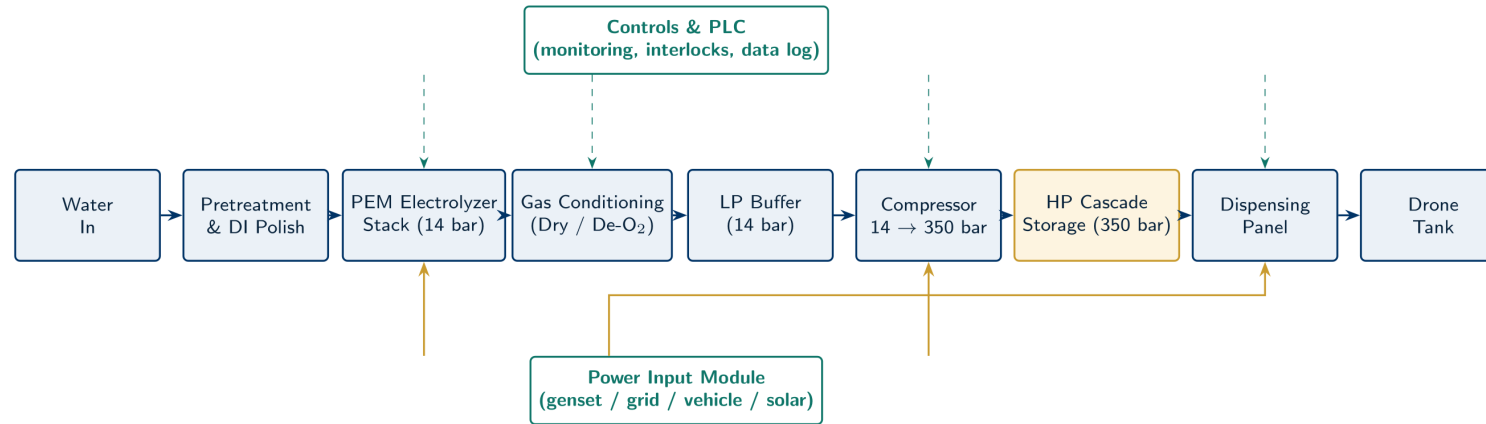
One big plant, trucks and pipes, depots, then users. Supply is someone else's, on someone else's schedule, priced for the whole chain.

Hydrodomus: decentralized

Water and electricity, electrolysis, compression, the user's tank. The user owns the supply, with no distribution chain.

We build compact, modular hydrogen generators that sit **next to the consumer**.

How it works



Generate

PEM electrolysis turns treated water and electricity into hydrogen, with no emissions on site.

Compress

Conditioning, compression and a buffer to the pressure the user's equipment needs.

Dispense

Direct fill into the end use, with safety interlocks and fail-closed valves on every stage.

Why decentralized production wins

Logistics disappear

Nothing to truck, swap or schedule. The hardest and costliest part of the hydrogen chain is removed.

Clean by design

Only water and electricity go in. Pair it with clean power and the hydrogen is clean from the first kilogram.

Autonomy

The operator controls supply, timing and resilience, with no dependency on a delivery network.

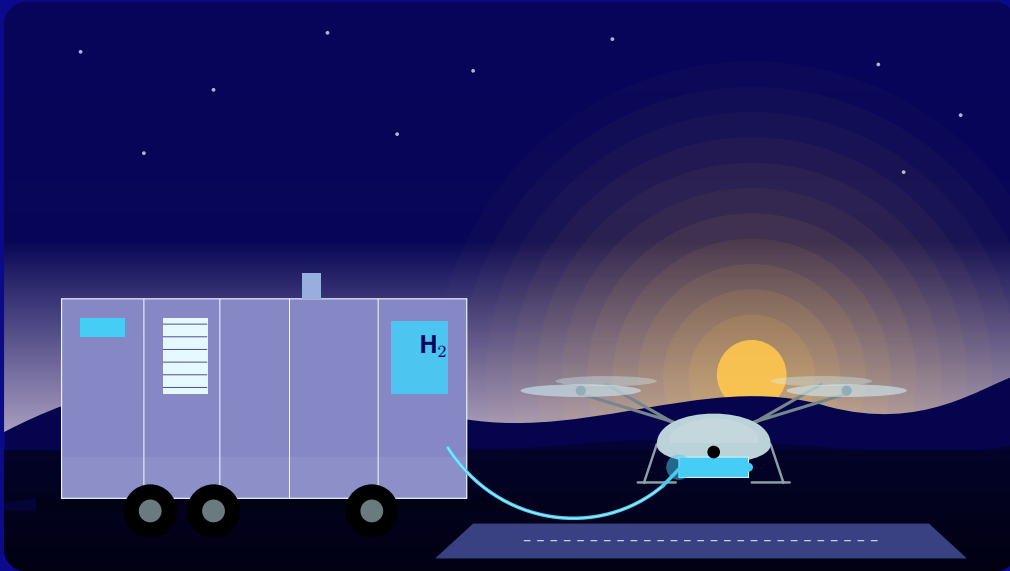
The value is greatest where **delivering hydrogen is hardest**.

Where we start, and where it leads

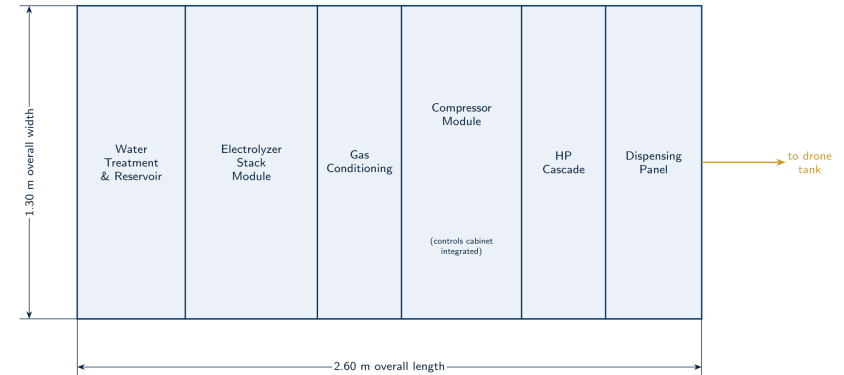
Stage	Segment	Why
Now	Drone fleets in the field (defense and dual-use)	Hydrogen drones are entering service and their fuel is still shipped in. Demand is urgent, logistics is the product.
Next, to validate	Remote and off-grid sites, backup power	Delivery is costly or impossible, on-site generation pays for itself.
Next, to validate	Fleet depots and industrial users	Predictable demand, a site that can host a unit.
Where we began	Home refueling for fuel-cell cars	Same core technology. We moved to defense where demand is more urgent and customers can commit sooner.

One technology core, applied first where decentralization is most valuable.

First product: Refuel Node TRN-10



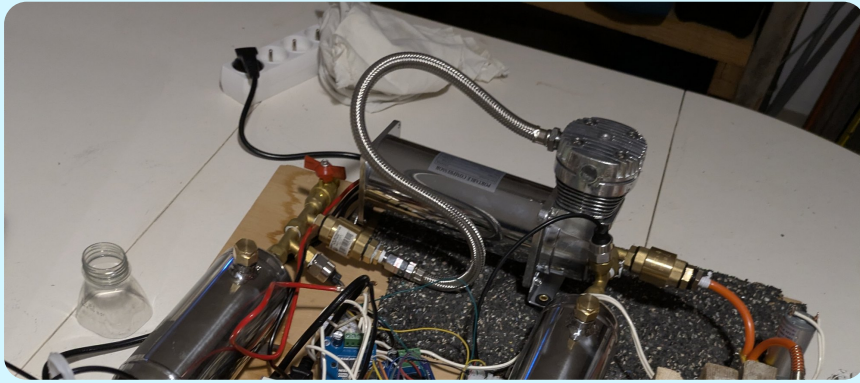
- Output** 10 kg H₂/day, modular to 25 kg/day
- Fill** 350 bar, under 5 min per tank
- Inputs** water and available power: generator, grid, vehicle or solar
- Form** trailer or skid, 2.6 × 1.3 × 2.05 m



General arrangement, top view

v1 design targets. The unit refuels hydrogen drones at the forward position, so no hydrogen is shipped in. Illustration is schematic.

Progress and honest status



In-house bench prototype: PEM electrolysis, filtration, two-stage compression.

1 of 3

target customers in discussion; meeting in October

11

supplier quotes in hand, from 20 requested

6 NL/h

working PEM bench electrolyzer

- Bench prototype proves core electrolysis know-how, about 13 g/day. The product target is roughly 770 times larger.
- Production stack and compressor are bought from established suppliers, not scaled up from the bench.
- Not yet: revenue, a signed letter of intent, a 350 bar fill, environmental testing.

Roadmap

1

Win the first customer

Convert the active customer into a letter of intent or pilot order. Firm up supplier offers and lock the pilot bill of materials.

2

Build the pilot

350 bar bench demo with a real tank interface, then ruggedization and testing with partners.

3

Field pilot

A unit at a customer site, generating and filling in real conditions.

4

Product family

Modular output tiers from the same core, then the next segments once the first is proven.

Frankfurt, February 2027: present pilot progress and test data.

Team



Yannick Durindel

Founder: business development and engineering

MSc Hydrogen Engine, University of Duisburg-Essen
ICAM engineering graduate, racecar engineering (MUNER)

Former developer at Eyalights, consultant at mc2i

Built the bench prototype solo, full-time on Hydrodomus

Advisors

David Holderbach

Hydrogen vehicles, founder of Hyvia.
Supplier and customer network.

Alysha Maria Lobo

Robotics and deep tech. Warm customer introductions.

The ask

€750K

at €7.5M post-money (10%)

Funds, in order of priority:

- Pilot unit build and integration
- Environmental and safety testing
- Certification and hazard studies
- Runway to deliver the above

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What the round delivers

1. A signed Lol or pilot order from the first customer
2. A 350 bar fill into a real drone tank
3. An environmental test report
4. A field pilot with real data

How you can help

Investors: introductions in deep tech, energy or defense tech.

Customers: operators who need hydrogen where supply is hard.

Grants: leads on non-dilutive funding for pilots.